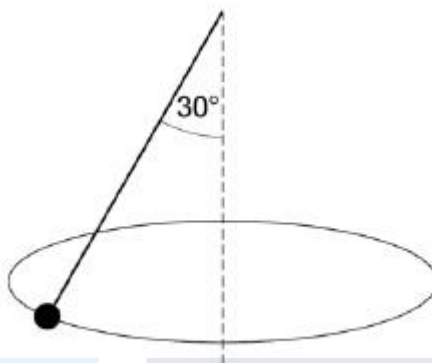


Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$.

A conical pendulum consists of a string of length 60 cm and a particle of mass 400 g. The string is at an angle of 30° to the vertical, as shown in the diagram.

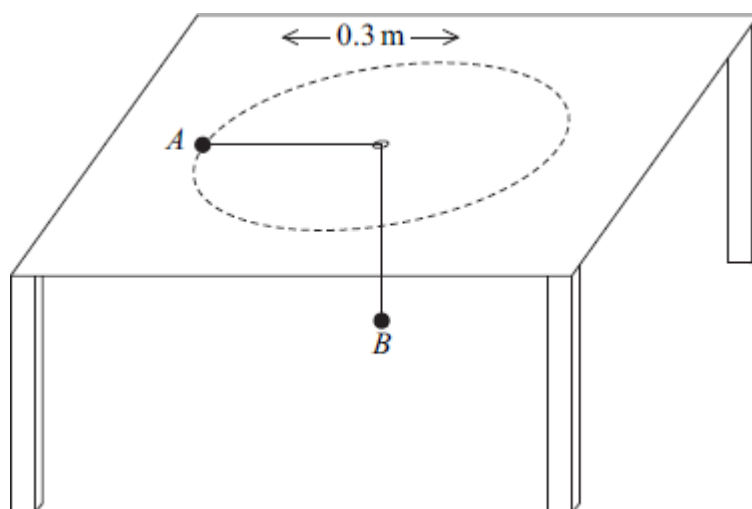


- (a) Show that the tension in the string is 4.5 newtons. (2)
- (b) Find the angular speed of the particle. (3)
- (c) State **two** assumptions that you have made about the string. (1)

(Total 6 marks)

Q2.

Two particles, *A* and *B*, are connected by a light inextensible string which passes through a hole in a smooth horizontal table. The edges of the hole are also smooth. Particle *A*, of mass 1.4 kg, moves, on the table, with constant speed in a circle of radius 0.3 m around the hole. Particle *B*, of mass 2.1 kg, hangs in equilibrium under the table, as shown in the diagram.



- (a) Find the angular speed of particle A . (4)
- (b) Find the speed of particle A . (2)
- (c) Find the time taken for particle A to complete one full circle around the hole. (2)
- (Total 8 marks)

Q3.

- (a) A shiny coin is on a rough horizontal turntable at a distance 0.8 m from its centre. The turntable rotates at a constant angular speed. The coefficient of friction between the shiny coin and the turntable is 0.3.

Find the maximum angular speed, in radians per second, at which the turntable can rotate if the shiny coin is not going to slide.

(4)

- (b) The turntable is stopped and the shiny coin is removed. An old coin is placed on the turntable at a distance 0.15 m from its centre. The turntable is made to rotate at a constant angular speed of 45 revolutions per minute.

- (i) Find the angular speed of the turntable in radians per second.

(2)

- (ii) The old coin remains in the same position on the turntable.

Find the least value of the coefficient of friction between the old coin and the turntable needed to prevent the old coin from sliding.

(4)

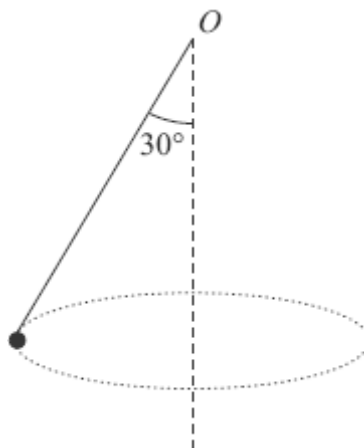
(Total 10 marks)

EXAM PAPERS PRACTICE

Q4.

©2025 Exam Papers Practice. All rights reserved.

A particle, of mass 6 kg, is attached to one end of a light inextensible string. The other end of the string is attached to the fixed point O . The particle is set in motion, so that it moves in a horizontal circle at constant speed, with the string at an angle of 30° to the vertical. The centre of this circle is vertically below O .

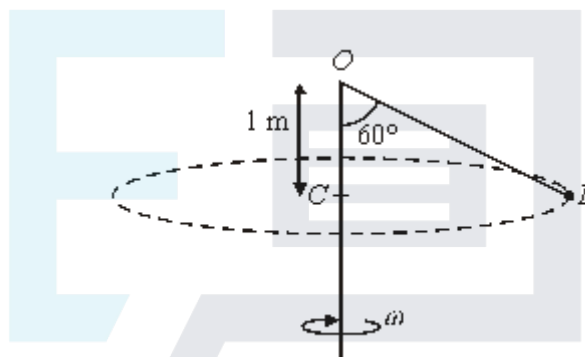


The particle moves in a horizontal circle with an angular speed of 40 revolutions per minute.

- (a) Show that the angular speed of the particle is $\frac{4\pi}{3}$ radians per second. (2)
- (b) Show that the tension in the string is 67.9 N, correct to three significant figures. (3)
- (c) Find the radius of the horizontal circle. (4)
- (Total 9 marks)

Q5.

The diagram shows a light inextensible cord that has one end O attached to the top of a fixed vertical pole.



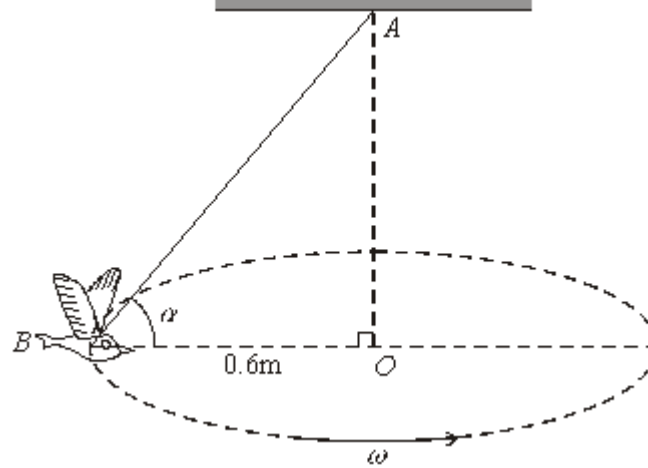
A ball B of mass 0.4 kg is attached to the other end of the cord. The ball rotates around the pole with constant angular speed ω radians per second. It moves in a horizontal circle with centre C at a distance 1 metre vertically below O . The angle between OB and OC is 60° .

- (a) Show that the tension in the cord is 7.84 newtons. (3)
- (b) Show that $\omega = 3.13$, correct to three significant figures. (5)
- (c) Show that the ball takes approximately two seconds to travel once around the pole. (1)
- (Total 9 marks)

Q6.

A toy bird, of mass 0.25 kg, is fixed to one end, B , of a light inextensible string. The other end, A , of the string is attached to a fixed point on a ceiling. The bird is set in motion, so that it describes a horizontal circle of radius 0.6 m. The centre of this circle is O , which is vertically below A , as shown in the diagram. The angle ABO is α .

The bird moves with a constant angular speed of ω radians per second.



The bird completes one full revolution every 1.5 seconds.

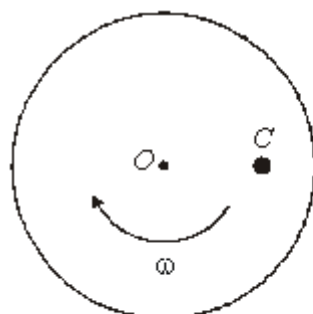
- (a) Find ω , leaving your answer in terms of π . (2)
- (b) (i) Determine the magnitude of the acceleration of the bird. (2)
- (ii) Show the direction of this acceleration on a diagram. (1)
- (c) (i) Draw a diagram to show the forces acting on the bird. (1)
- (ii) Find the angle α , giving your answer to the nearest degree. (6)
- (d) State **one** modelling assumption used in this question. (1)

(Total 13 marks)

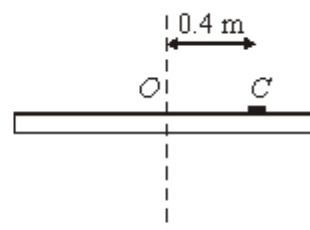
©2025 Exam Papers Practice. All rights reserved.

Q7.

A small coin, C, is placed on a rough horizontal turntable, as shown in the diagram.



Plan view



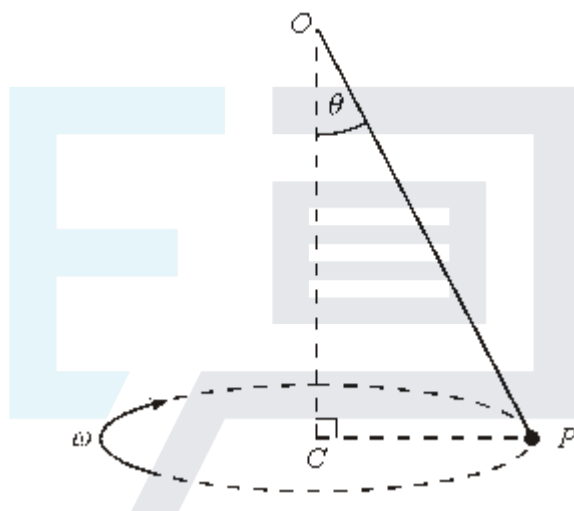
Side view

The mass of the coin is 0.01 kg and it is placed at a distance of 0.4 m from O, the centre of the turntable.

The turntable rotates about a vertical axis through O , with a constant angular speed of ω radians per second. During this motion the coin does **not** slip.

- (a) Draw a diagram to show all the forces acting on the coin. (1)
 - (b) Find, in terms of ω , an expression for the frictional force acting on the coin. (2)
 - (c) The coefficient of friction between the coin and the turntable is 0.8. The coin is on the point of slipping. Show that the value of ω is approximately 4.43. (4)
- (Total 7 marks)**

Q8.



A particle, P , of mass m is attached by a light, inextensible thread of length l to a fixed point O . The particle moves with constant angular speed ω in a horizontal circle with centre C . The point C is vertically below O . The thread is inclined to the vertical at an angle θ , as shown in the diagram.

- (a) Find an expression for the tension in the thread. Give your answer in terms of m , l and ω . (4)
- (b) Show that $\cos \theta = \frac{g}{l\omega^2}$. (3)
- (c) The greatest tension that the thread can withstand without breaking is 16 N.
In the case when $l = 0.4$ metres and $m = 0.1$ kg:
 - (i) show that $\omega \leq 20$ radians per second; (2)
 - (ii) find the greatest possible value of θ , giving your answer to the nearest degree.

(2)
(Total 11 marks)

Q9.

A spin drier is tested when empty. The drum, initially at rest, reaches its maximum angular speed of 1 100 revolutions per minute in 8 seconds.

- (a) Express the maximum angular speed of the drum in radians per second.
- (b) Assuming that the angular acceleration of the drum is constant, calculate its magnitude.

(2)
(Total 4 marks)

Q10.

The orbit of the moon around the earth may be modelled as circular. The time taken for the moon to make one complete orbit is approximately 27.32 days.

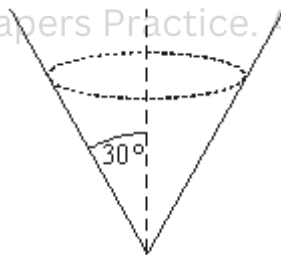
- (a) Show that the angular speed of the moon is approximately 2.66×10^{-6} radians per second.
- (b) Assuming that the radius of the circular orbit is approximately 3.844×10^8 metres, find the speed of the moon, relative to the earth, in metres per second.

(2)
(Total 5 marks)

Q11.

A particle, of mass 3 kg, describes a horizontal circular path on the inside surface of a smooth cone, as shown in the diagram.

©2025 Exam Papers Practice. All rights reserved.



The radius of the circle is 0.5 metres and the semi-vertical angle of the cone is 30° . The particle moves at a constant speed.

- (a)
 - (i) Show that the magnitude of the normal reaction force on the particle is 58.8 N.
 - (ii) Find the speed of the particle.
- (b) The particle moves on the same cone, but in a horizontal circle of greater radius than before.

(i) What happens to the magnitude of the normal reaction force?

(1)

(ii) What happens to the speed of the particle? Explain your answer.

(2)

(Total 10 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.